Robotics and Vision

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Robotics and Vision (A13362)

Short Title: Robotics and Vision
Faculty Science and Computing
Department Engineering Technology
Cluster Automotive, Automation and IoT
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Credits 5 **Level:** Advanced

Description of Module / Aims

This module introduces the learner to the field of robotics. The mechanical construction and control of a range of robot types is explored, together with the suitability of the systems to industrial application. The learner will utilise industrial robots to explore the various module topics along with various off-line programming packages to develop programming competencies. Topics in vision processing and analysis are evaluated and implemented using vision processing software.

Programmes

`ROBS-0001`	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	4 / 8 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	4 / 8 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	4 / 8 / M
`ROBS-0001`	BEng (Hons) (`WD_ETRIC_B`)	4 / 8 / M

Indicative Content

- Physical Systems on Robotics: Manipulator configurations and work envelopes; degrees of freedom.
- Robot programming - offline; on-line; co-ordinate systems - World, Base, Joint, Tool
- Control Systems: Control Strategies; Co-ordinate systems; Robot performance.
- Vision systems: Image capture and processing using commercial software and hardware.
- Vision processing: thresholding, image enhancement, smoothing, siltering, edge detection. Colour and grayscale techniques.
- Safety: Operator safety, teach mode, speed control, safe zones

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Differentiate between a range of robot manipulator configurations and control strategies and discuss the performance of each.
2. Evaluate the suitability of robots for particular applications.
3. Produce, download and test software programs to control industrial robots.
4. Utilise 'offline' programming software to develop and debug programmes and control strategies for industrial robots.
5. Demonstrate an ability to work safely on engineering equipment without ever threatening to endanger themselves or others.
6. Develop and implement standard vision processing techniques in software.
7. Design and implement a practical vision system for an engineering application.

Learning and Teaching Methods

- Lectures

- Practical laboratory work
- Mini project

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
In-Class Assessment	25%	1,2,3,4,5
Practical	25%	3,4,5
In-Class Assessment	30%	6,7
Project	10%	6,7
Practical	10%	6,7

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Essential Material

- A.B.B. Robotics. _Operating Manual RobotStudio_ Sweden. 2015.
- Glaser, A. _Industrial Robotics: How to implement the right system for your plant_. UK: Industrial Press, 2009.
- Staubli Faverges. _Staubli Rx90 V+ Programming Manual_ .. 2004.
- Turner, I.C. _Engineering applications of pneumatics and hydraulics_. UK: Arnold: Butterworth heinemann, 2014.

Supplementary Material

- Saha, S.K. _Introduction to Robotics_. .: McGraw-Hill, 2014.

Requested Resources

- Room Type: Computer Lab